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Short Term Cert Developer Manual

Version: 1.1
Date: 19th February 2026

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Contents

1 Overview **5**
1.1 Ident Record service . . . 5
1.2 Order of operations . . . 5
1.3 Quick reference - Short Term Cert issuing and signing service . . . 6
1.4 Generic return codes . . . 6
1.5 Timeouts . . . 6
1.6 Signature format . . . 6

2 API description **8**
2.1 Issue . . . 8
2.1.1 Request . . . 8
2.1.2 Response . . . 8
2.2 Get certificate . . . 9
2.2.1 Request . . . 9
2.2.2 Response . . . 9
2.3 Start signature . . . 9
2.3.1 Request . . . 10
2.3.2 Response . . . 10
2.4 Get signature . . . 11
2.4.1 Request . . . 11
2.4.2 Response . . . 11
2.5 Get session info . . . 11
2.5.1 Request . . . 11
2.5.2 Response . . . 12
2.6 Close session . . . 12
2.6.1 Request . . . 12
2.6.2 Response . . . 12

3 Test System **13**

References **14**

Änderungsverlauf

Date	Rev	Author	Changes
19.02.2026	1.1	Patrick Hagelkruys	Update description Section test system
29.01.2026	1.0	Patrick Hagelkruys	Update product name
08.01.2026	0.3	Alexander Kopp Joel Chinnow Patrick Hagelkruys	Review, optimize rest calls remove duplicate calls
05.01.2026	0.2	Patrick Hagelkruys Ramin Sabet	Review, process description additional rest calls
02.01.2026	0.1	Patrick Hagelkruys	first version

Table 1: Document history

1 Overview

This document describes the API interface for issuing Short Term Cert certificates and for signing with those certificates. Figure 1 shows the general workflow.

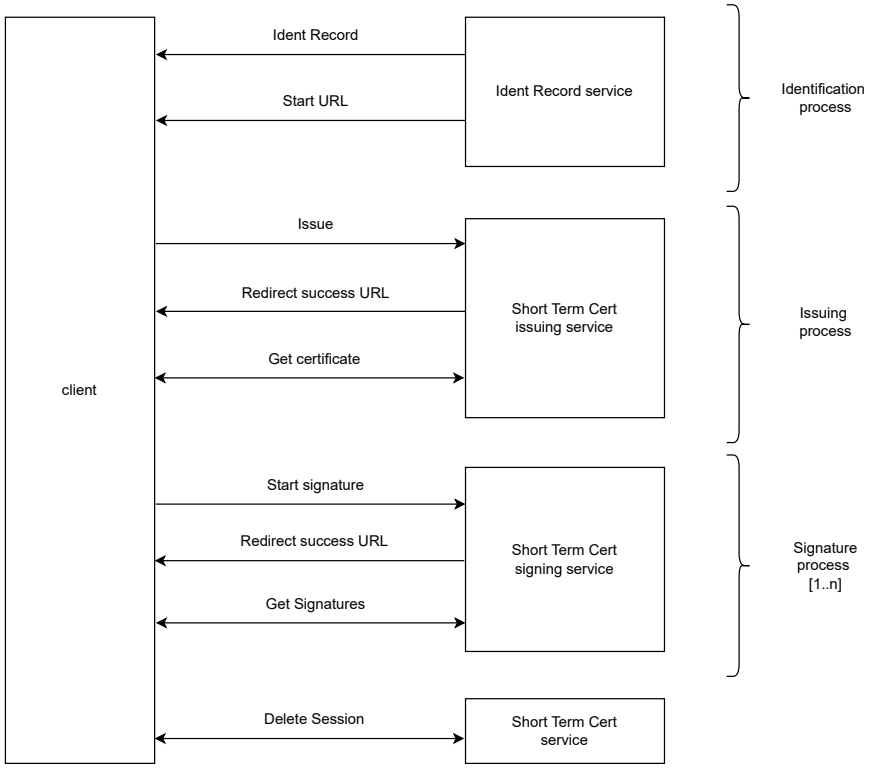

Figure 1: Workflow

1.1 Ident Record service

The Ident Record service is described in [HS]. To use the interface, an authentication certificate is required, which can be obtained from A-Trust after the legal contract has been concluded.

Further information is available via `https://github.com/A-Trust/IdentInterface`.

1.2 Order of operations

As shown in Figure 1, the first step is identification, i.e., calling the Ident Record service.

The next step is issuing the certificate.

After successful certificate issuance, the signing process can be called as many times as needed.

Once all signatures have been completed, it is strongly recommended to use the „Delete Session“ command to end the workflow.

1.3 Quick reference - Short Term Cert issuing and signing service

URL	Method	Functionality
issue/<token>	POST	start the issuing of the Short Term Cert
certificate/<ticketId>	GET	get the issued certificate
signature/<ticketId>	POST	add hash to be signed and start signature
signature/<ticketId>	DELETE	get all signature and delete document from server
session/<ticketId>	GET	retrieves session information, such as validity.
session/<ticketId>	DELETE	deletes the session and data

Table 2: Quick reference

1.4 Generic return codes

The following return codes can be returned with any REST call. Therefore, they are not explicitly listed.

HTTP/500 internal server error: An error occurred while processing the request.

HTTP/400 bad request: There is an error in the request.

HTTP/401 Unauthorized: There session timeout is reached.

1.5 Timeouts

There is a 15-minute timeout between calls to the services.

1.6 Signature format

Certificates for Short Term Cert are issued with an ECDSA-256 private key. Signing is performed using SHA-256 hash.

The signature field hast the following format:

$$
\begin{aligned}
signature &= Base64(\ R + \ S) \\
\text{where } R &= \text{first coordinate of ECDSA point} \\
S &= \text{second coordinate of ECDSA point}
\end{aligned}
$$

2 API description

2.1 Issue

The IdentRecord service described in [HS] returns this URL. The parameter `token` is also returned from the IdentRecord service.

In response to a HTTP POST to this URL, a `ticketId` and a `startUrl` are returned.

The `ticketId` is used in subsequent requests.

The user should be redirected to `startUrl` to accept the legal documents and issue the certificate.

After the certificate is issued, the user (browser) is redirected to the `redirectUrl` or, in case of an error, to the `errorUrl`.

2.1.1 Request

```
POST <path>/api/v1/Issue/<token> HTTP/1.1
Host: ...
Content-Length: 260
Content-Type: application/json; charset=utf-8

{
  "redirectUrl": "https://www.app.at/success",
  "errorUrl": "https://www.app.at/error"
}
```

Listing 1: Issue request

2.1.2 Response

```
HTTP/1.1 200 OK
Content-Length: 130
Content-Type: application/json; charset=utf-8

{
  "ticketId": "AABSDKLAFKJAKJSD",
  "startUrl": "https://..../ASD1247187178287312"
}
```

Listing 2: Issue response

2.2 Get certificate

Once the certificate has been issued according to the request in the chapter 2.1, the issued certificate can be downloaded via this URL.

2.2.1 Request

```
GET <path>/api/v1/certificate/<ticketId> HTTP/1.1
Host: ...
```

Listing 3: Get certificate request

2.2.2 Response

```
HTTP/1.1 200 OK
Content-Type: application/pkix-cert
Content-Length: 1023

<binary data>
```

Listing 4: Get certificate response

2.3 Start signature

For signing, the hash values (`hashes`) must be uploaded. These also contain a document name (`name`) for display and a link (`uri`) to the original document so the user can view it.

The response contains a `startUrl`, the generated document IDs (`docId`) alongside the document name (`name`), which are later used to download the signature. The (`docId`) are in the same order as the uploaded hash values.

The user should be redirected to `startUrl` to sign the documents.

After the signature process is complete, the user (browser) is redirected to the `redirectUrl` or, in case of an error, to the `errorUrl`.

2.3.1 Request

```
POST <path>/api/v1/signature/<ticketId> HTTP/1.1
Host: ...
Content-Length: 1612
Content-Type: application/json; charset=utf-8

{
  "redirectUrl": "https://www.app.at/success",
  "errorUrl": "https://www.app.at/error",
  "hashes": [
    {
      "hash": "AAasgj...kjf=",
      "name": "Document Name",
      "uri": "https://www.documenturl.at/afhak"
    },
    ...
    {
      "hash": "Hdipol...loa=",
      "name": "Document Name 2",
      "uri": "https://www.documenturl.at/htre"
    }
  ]
}
```

Listing 5: Start signature request

2.3.2 Response

```
HTTP/1.1 200 OK
Content-Length: 962
Content-Type: application/json; charset=utf-8

{
  "startUrl": "https://..../ASD1247187178287312",
  "docIds": [
    {
      "name": "Document Name",
      "id": "0d4e001c-d04a-4596-a808-c6b70cdbd929"
    },
    ...
    {
      "name": "Document Name 2",
      "id": "6798321c-eab9-48ec-92f2-8358a9b5228c"
    }
  ]
}
```

Listing 6: Start signature response

2.4 Get signature

Download all available signature, including the `docId` for reference. The hash and signature values are deleted from the server after downloading.

2.4.1 Request

```
DELETE <path>/api/v1/signature/<ticketId> HTTP/1.1
Host: ...
```

Listing 7: Get signature request

2.4.2 Response

```
HTTP/1.1 200 OK
Content-Length: 678
Content-Type: application/json; charset=utf-8

{
  "signatures": [
    {
      "docId": "0d4e001c-d04a-4596-a808-c6b70cdbd929",
      "signature": "ABsdjkhaut1..1223=",
    },
    ...
    {
      "docId": "6798321c-eab9-48ec-92f2-8358a9b5228c",
      "signature": "HjkLKFASasd ...tzia=",
    }
  ]
}
```

Listing 8: Get signature response

2.5 Get session info

Request to get the validity of the current session and the validity of the private key.

2.5.1 Request

```
GET <path>/api/v1/session/<ticketId> HTTP/1.1
Host: ...
```

Listing 9: Get session info request

2.5.2 Response

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: 1023

{
  "sessionValidTo": "2026-01-05T12:58:17.1975174Z",
  "certificate": {
    "data": "",
    "validFrom": "2026-01-05T12:47:17.1975174Z",
    "validTo": "2026-01-05T13:47:17.1975174Z"
  },
  "sigKeyValidTo": "2026-01-05T12:58:16.1975174Z"
}
```

Listing 10: Get session info response

2.6 Close session

HTTP request to delete the session. Can be called at any time to cancel the transaction or end the session and prevent future signatures.

2.6.1 Request

```
DELETE <path>/api/v1/session/<ticketId> HTTP/1.1
Host: ...
```

Listing 11: Close session request

2.6.2 Response

```
HTTP/1.1 200 OK
```

Listing 12: Close session response

3 Test System

For testing purpose the following system ist provided:

URL https://hs-abnahme.a-trust.at/einmalzertifikate/api/v1

References

[HS] Hagelkruys, Patrick and Ramin Sabet: *Description identity interface and identity record)*. https://github.com/A-Trust/Identschnittstelle.